

KALI: A Lightweight Custom Linux Distribution & System Management Suite

CIT 322 :: Operating System Sessional

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1. Introduction

This project demonstrates the practical application of Operating System principles through the development of a custom, lightweight Linux distribution named KALI, alongside a suite of system administration and process monitoring tools. KALI is built by customizing the ArchLinux ISO [3] to create a bare-minimum live environment populated only with essential modern applications. It utilizes archinstall for a streamlined, script-free base installation, ensuring a lightweight footprint. The environment is graphical-ready, customized with the Hyprland Wayland compositor, SDDM, Waybar, Rofi, and Waypaper.

To bridge the gap between low-level system configuration and user-space process management, complementary tools were developed: Archmate for comprehensive system administration, a Chaotic AUR installer script, and Tree_tasker, a Python and PySide6-based graphical task manager with web integration for advanced process hierarchy visualization. For automated setups, dotfiles can be integrated, as well as third party providers like, END4 [1], ML4W [2] scripts.

2. Objectives

1. To understand Linux system architecture by customizing and building a bootable Arch-based ISO[3], a rolling based distributions.
2. To customize a window manager with lua scripts and custom shell scripts.

3. To automate core OS maintenance tasks including package, user, repository, and configuration management.
4. To visualize and monitor system resource allocation (CPU, RAM, Disk, Network) per application in real-time.
5. To map and interactively explore parent-child process relationships (PIDs) within the operating system.
6. To allow users to use remote connections over ssh using terminal or vscode to remotely edit the operating system without any GUI.

3. Problem Statement

Modern operating systems often come pre-packaged with heavy, unnecessary background services (bloatware) that consume valuable hardware resources. Furthermore, traditional command-line process monitors (like top or ps) provide dense tabular data that makes it difficult for users to intuitively understand parent-child process relationships and exact resource distribution. There is a need for a streamlined, customizable OS base paired with an intuitive, visual process manager that clearly maps process hierarchies and resource bottlenecks.

4. Related Work

1. General-purpose Linux distributions (e.g., Ubuntu [4], Linux Mint [5]) provide functional environments but lack the minimalism required for strict resource control.
2. ArcoLinux provides minimal functionality but the project is dead now [6].
3. Popular distributions delay to release applications [7], which can be crucial for development.

5. Methodology

OS Building

1. Kernel: Linux-7.0.12 (amd64)
2. Userspace: GNU
3. Package Management: Pacman, AUR

System Tools

1. OS Customization: Archiso builder, archinstall
2. UI: Waybar, Rofi, Waypaper
3. Scripting & Administration: Bash scripting (Archmate, Chaotic AUR Installer)

6. Design Principles

1. The design principle is to keep the core OS live ISO as small as possible, relying on native tools.
2. To separate system administration from real-time process monitoring.

7. Limitations

1. Currently this project is built only for x86_64. 32 bit or ARM CPU won't work on it.
2. Wayland may not work properly on NVIDIA GPU with some window managers [8].

8. Conclusion

We can conclude that our project results in a fully functional operating system, installable via custom user scripts or archinstall from official Arch repository. Finally, the Tree_tasker application accurately polls kernel data to display real-time parent-child process trees and accurately renders CPU, RAM, Disk, and Network usage into interactive pie charts across both desktop and web interfaces.

9. References

1. <https://github.com/end-4/dots-hyprland>
2. <https://github.com/end-4/dots-hyprland>
3. <https://archlinux.org/>
4. <https://kali.com/>
5. <https://www.linuxmint.com/>
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